Data-Backed Journal-Style Replication of *Ambition Is Nothing Without Focus*

Replication Paper Project

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Abstract

This paper presents a data-backed replication of Ambition Is Nothing Without Focus: Compensating for Negative Transfer of Experience in R&D using only processed datasets and result files available in the workspace. The replication preserves the source paper's core empirical logic while making every local substitution explicit. The paper is therefore reported as near-structure-preserving local reproduction rather than as a source-data replay. The current local batch contains 489,193 observations across 200 entities, and the batch headline estimate is -0.0022 for Breadth x Org Recomb Exp log with p = 0.008. The local rerun strongly preserves the negative breadth-by-experience interaction and the negative three-way interaction with organizational focus. The contribution is a journal-style replication manuscript that separates direct source claims from executable local evidence.

Keywords: recombination breadth; organizational focus; inventive impact; patent citations; local reproduction

1 Introduction

This paper begins from a paper-first reading of Ambition Is Nothing Without Focus: Compensating for Negative Transfer of Experience in R&D and treats the source article as an empirical argument that has to be rebuilt before it can be judged. The present manuscript is classified as near-structure-preserving local reproduction. A paper-first restatement of the empirical question is as

follows: The source paper asks whether broad recombination becomes less productive when it leans on familiar routines, unless organizational focus offsets that negative transfer (Ghosh et al., 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The replication matters because it targets a mechanism-heavy paper where the interaction structure, not only the sign of a main effect, is the core theoretical claim. The original article does not rest on a descriptive regularity alone. It ties a theory about search, differentiation, timing, or organizational learning to a specific unit of analysis, sample frame, and estimand. A careful replication therefore has to recover the structure of that logic before any local coefficients are interpreted (Ghosh et al., 2014; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The executable evidence base used in this manuscript contains 489,193 observations across 200 entities and spans 1986 to 2016 in the local executable panel. Those numbers are not filler. They define the actual inferential scope of the local reconstruction and set the boundary on what can be claimed honestly from the available datasets (Ghosh et al., 2014; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The paper's contribution is a near-structure-preserving reproduction that shows how the source mechanism survives in a fully local cleaned patent panel. The paper is written to read like a formal journal article rather than an internal workflow memo. The main text develops the theoretical logic, data reconstruction, derivations, empirical design, and results in continuous prose, while the appendices preserve the audit trail required for reruns and later extensions (Ghosh et al., 2014; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The paper belongs to a top-journal literature on exploration, recombination, and the organizational conditions under which broad search helps or hurts inventive performance. (Ghosh et al., 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Cohen and Levinthal, 1990; Kogut and Zander, 1992, 1996; Teece et al., 1997; Benner and Tushman, 2003). That literature matters because the replication is not asking whether breadth is always good; it is asking whether the interaction between breadth, organizational experience, and focus survives in a fully executable patent panel. (Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Henderson and Clark, 1990; Anderson and Tushman, 1986; Hargadon and Sutton, 1997; Tripsas and Gavetti, 2000; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Stuart, 2000). The present manuscript therefore treats the source paper as one contribution inside a broader top-journal stream on exploration, recombination,

performance measurement, alliance-based learning, inventive search, technological discontinuity, and patent-based inference. The literature review is intentionally broad because each replication paper should stand on a serious journal-level reference base rather than on the source article alone. (Rosenkopf and Almeida, 2003; Schilling and Phelps, 2007; Almeida and Kogut, 1999; Fleming et al., 2007; Jaffe, 1986; Jaffe et al., 1993; Trajtenberg, 1990; Hall et al., 2005; Henderson and Cockburn, 1994). For this manuscript, the reference base extends further into top-journal work on organizational design, temporal exploration, alliance-based exploration-exploitation balance, and the mapping role of science in technological recombination because those papers speak directly to why broad search can become productive in some organizational settings and harmful in others. (Fleming and Sorenson, 2004; Nerkar, 2003). A second topic-specific literature block is added to sharpen the manuscript's empirical boundary and identification problem. (Rivkin and Siggelkow, 2003; Lavie et al., 2011).

2 Original Study, Theory, and Replication Boundary

The focal source article is (Ghosh et al., 2014). The original study can be summarized as a mechanism paper with a sharp empirical core. The source paper studies the interaction of breadth, organizational experience, and focus. Forward citations proxy inventive impact. The unit of analysis is the patent. That combination matters because the source article asks readers to believe not merely that two variables move together, but that a particular organizational or strategic mechanism explains why the relationship should appear in the data (Ghosh et al., 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). In a setting like this, faithful replication requires more than reusing a title and rerunning a regression. The theoretical contribution lives in the way constructs are defined, how they enter the estimand, and which comparison the model is supposed to adjudicate. For that reason, this manuscript first reconstructs the published theory and only then turns to the local executable analogue (Ghosh et al., 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The replication boundary is therefore explicit. Direct source facts are carried over when they can be read from the published article itself. Local replication decisions appear only when the current workspace must translate the source design into verified processed

datasets. The goal is disciplined mechanism recovery, not rhetorical imitation that quietly changes the design (Ghosh et al., 2014; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996).

2.1 Original Paper Methodology

A faithful evaluation of Ghosh, Martin, Pennings, and Wezel (2014) starts with the published empirical design rather than with the local rerun. The source article studies patent-level inventive impact in firm R&D and centers its theory on a structured interaction among recombination breadth, organizational recombination experience, and organizational focus (Ghosh et al., 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The published sample is described as follows: The published design is a patent-level study of firm R&D using an archival panel of assigned patents and firm controls. The published measurement architecture is described as follows: The published analysis uses patent-level inventive impact and interaction terms among breadth, organizational experience, and organizational focus. These details matter because they determine what the published paper actually observes rather than what a later analogue can observe (Ghosh et al., 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The published estimator is described as follows: patent-level count models with interactions among breadth, organizational recombination experience, and organizational focus, together with patent controls and firm-year structure. The crucial published comparison is whether the marginal value of broad recombination changes once the same firm has accumulated more organizational experience and whether focus changes that transfer pattern. Reading the source methodology this way prevents the current manuscript from quietly replacing the original design with one generic fixed-effects story (Ghosh et al., 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993).

2.2 Replication Status and Boundary

Replication status for this paper: Near-structure-preserving local reproduction (Ghosh et al., 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). This paper is the closest thing in the current archive to a near-structure-preserving local reproduction. The executable data stack preserves the patent-level unit, the key interaction structure, and the inventive-impact outcome,

even though exact sample filters remain local rather than archival-source identical (Ghosh et al., 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993).

2.3 Direct Source Facts Versus Replication Decisions

The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that The interaction structure is the theoretical core. The local reconstruction therefore adopts the following bounded decision: The local rerun keeps the same interaction logic on the clean GVKEY-based panel. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline (Ghosh et al., 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that Forward citations measure impact. The local reconstruction therefore adopts the following bounded decision: The local impact measure is the executable citation outcome already stored in the panel. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline (Ghosh et al., 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that The paper works at the patent level. The local reconstruction therefore adopts the following bounded decision: The sample is narrowed to active S&P 500 utility patents with unambiguous assignment. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built

from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline (Ghosh et al., 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993).

3 Data, Sample, and Measures

The local evidence chain is deliberately modular. The source paper is treated as the authority for the research question, theoretical mechanism, and published design. Estimation, by contrast, proceeds only through the verified local files. The required dataset types in this paper are Clean GVKEY-anchored S&P 500 utility-patent panel from the local replication pipeline. PatentView-derived application year and citation outcomes. CRSP-Compustat linked firm-year filters and sector information already present in the workspace (Ghosh et al., 2014; Jaffe, 1986; Hall et al., 2005). Operationally, the reconstruction relies on four evidence layers: the analysis panel, the coefficient export, the sample summary, and, when available, the fit summary. The resulting executable panel supplies 489,193 observations across 200 entities and covers 1986 to 2016 in the local executable panel. Every numeric claim in the results section is intended to trace back to one of those verified evidence layers rather than to a hand-copied number from the article (Ghosh et al., 2014; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996). This data-lineage discipline matters for two reasons. First, it prevents the manuscript from smuggling unverified preprocessing steps into the design. Second, it forces every substitution to be visible. If a construct cannot be reproduced from the current processed datasets, the paper states the analogue directly instead of pretending that the original archive has been fully recreated (Ghosh et al., 2014; Jaffe, 1986; Hall et al., 2005; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The local rerun uses the clean GVKEY-anchored utility-patent panel already built in the workspace and keeps the patent-level interaction structure of the original paper. The local sample frame is Active S&P 500 utility patents with unambiguous GVKEY assignment in the local cleaned reproduction pipeline The observed time window in the executable panel is 1986 to 2016 in the local executable panel The outcome is Patent-level inventive impact measured from forward citations (Ghosh et al., 2014; Jaffe, 1986; Hall et al., 2005). The main text works from a verified analysis panel, a verified coefficient export, and a verified sample summary. Exact

file names are retained in the replication notes and appendices, while the article body focuses on data lineage and empirical function (Ghosh et al., 2014; Hall et al., 2005).

3.1 Local Data Sources Used in This Replication

This replication uses a clean GVKEY-anchored patent-level reproduction panel, PatentView and USPTO-derived application-year and citation outcomes, and CRSP-Compustat-linked firm filters and controls already built into the local pipeline (Ghosh et al., 2014; Jaffe, 1986; Hall et al., 2005). Those local dataset families are not interchangeable: clean gvkey-anchored patent-level reproduction panel is used to define breadth, organizational experience, organizational focus, and the patent-level sample; patentview and uspto-derived patent outcomes is used to supply application timing and inventive-impact outcomes; crsp-compustat-linked firm filters and controls is used to enforce the executable public-firm sample and firm-level controls. Together, they support 489,193 observations across 200 entities over 1986 to 2016 in the local executable panel (Ghosh et al., 2014; Jaffe, 1986; Hall et al., 2005). This local data stack is narrower than the original article's data architecture. Clean GVKEY-anchored S&P 500 utility-patent panel from the local replication pipeline. PatentView-derived application year and citation outcomes. CRSP-Compustat linked firm-year filters and sector information already present in the workspace. The manuscript therefore keeps the boundary visible: the source paper supplies the original methodology, whereas the local processed dataset families and calculated outputs supply the estimable analogue. Exact filenames are retained in the replication notes rather than printed through the body text (Ghosh et al., 2014; Jaffe, 1986; Hall et al., 2005).

DatasetFamily	RoleInPaper
Clean GVKEY-anchored patent-level reproduction panel	define breadth, organizational experience, organizational focus, and the patent-level sample
PatentView and USPTO-derived patent outcomes	supply application timing and inventive-impact outcomes
CRSP-Compustat-linked firm filters and controls	enforce the executable public-firm sample and firm-level controls

Local data families used in the present paper.

3.2 Sample Reconstruction

The executable sample is constructed in three observable stages. The raw regression dataset begins with 499,486 patent-level observations. After the active-firm and patent-type merge, the panel still contains 499,486 rows, but 3,678 patents remain ambiguously assigned and therefore cannot be used for a clean firm-level reproduction. The final analysis sample retains 489,193 observations after restricting attention to active S&P 500 utility patents with unambiguous GVKEY assignment (Ghosh et al., 2014; Jaffe, 1986; Hall et al., 2005). This sample-construction sequence is important for interpretation. The final panel is not simply a large patent file; it is the outcome of a specific replication decision to privilege clean firm assignment over broader but noisier coverage. That decision preserves the organizational mechanism studied in the source paper, even though it may narrow the empirical frame relative to the original archive (Ghosh et al., 2014; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996).

Step	Rows	UniqueGVKEY	UniqueYears
regression_dataset_raw	499,486	204	31
after_active_and_patent_type_merge	499,486	204	31
sp500_utility_drop_ambiguous	489,193	200	31

Sample reconstruction from the current local dataset.

3.3 Variable Construction and Derivation

The construct Inventive impact is operationalized locally as Forward citations and appears in the executable dataset as inventive_impact. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Use the local cleaned citation outcome already stored in the regression dataset.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms,

this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Ghosh et al., 2014; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The construct Breadth is operationalized locally as Centered breadth and appears in the executable dataset as breadth_c. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Carry forward the centered breadth variable from the local regression output.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Ghosh et al., 2014; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The construct Organizational experience is operationalized locally as Centered log prior organizational experience and appears in the executable dataset as org_recomb_exp_log_c. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Carry forward the centered log experience variable from the same panel.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Ghosh et al., 2014; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The construct Organizational focus is operationalized locally as Centered firm-year focus and appears in the executable dataset as org_focus_c. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Use the contemporaneous focus measure already reconstructed in the local pipeline.. This mapping is retained because it preserves

the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Ghosh et al., 2014; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The construct Controls is operationalized locally as Prior-art citations, degree, field experience, scale, age and appears in the executable dataset as n_prior_art_cites; recomb_degree; field_recomb_exp_log; total_patents_same_year; firm_age. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Reuse the local patent-level controls and fixed effects.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Ghosh et al., 2014; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005).

Construct	Local measure	Derivation
Inventive impact	Forward citations	Use the local cleaned citation outcome already stored in the regression dataset.
Breadth	Centered breadth	Carry forward the centered breadth variable from the local regression output.
Organizational experience	Centered log prior organizational experience	Carry forward the centered log experience variable from the same panel.
Organizational focus	Centered firm-year focus	Use the contemporaneous focus measure already reconstructed in the local pipeline.

Construct	Local measure	Derivation
Controls	Prior-art citations, degree, field experience, scale, age	Reuse the local patent-level controls and fixed effects.

Variable mapping and derivation logic in the main text.

3.4 Construct-Level Derivation Logic

The derivation appendix records how each major construct enters the executable estimand for Ghosh, Martin, Pennings, and Wezel (2014). The point is not to rehearse notation for its own sake, but to show how the source paper's conceptual vocabulary is translated into verified columns in the local panel. For Inventive impact, the manuscript uses Forward citations under the exact local implementation inventive_impact. The derivation step is: Use the local cleaned citation outcome already stored in the regression dataset.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Breadth, the manuscript uses Centered breadth under the exact local implementation breadth_c. The derivation step is: Carry forward the centered breadth variable from the local regression output.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Organizational experience, the manuscript uses Centered log prior organizational experience under the exact local implementation org_recomb_exp_log_c. The derivation step is: Carry forward the centered log experience variable from the same panel.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Organizational focus, the manuscript uses Centered firm-year focus under the exact local implementation org_focus_c. The derivation step is: Use the contemporaneous focus measure already reconstructed in the local pipeline.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Controls, the manuscript uses Prior-art citations, degree, field experience, scale, age under the exact local implementation n_prior_art_cites; recomb_degree; field_recomb_exp_log; total_patents_same_year; firm_age. The

derivation step is: Reuse the local patent-level controls and fixed effects.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept.

4 Descriptive Analysis of the Current Dataset

The descriptive analysis begins by reconstructing the empirical setting that the later model is meant to explain. For Inventive impact, the mean is 6.225, the standard deviation is 22.91, and the interquartile range runs from 1.000 to 6.000; For Breadth, the mean is -0.0064, the standard deviation is 5.276, and the interquartile range runs from -3.050 to 0.950; For Organizational experience, the mean is 0.0070, the standard deviation is 1.346, and the interquartile range runs from -0.778 to 0.320; For Organizational focus, the mean is -0.00036, the standard deviation is 0.0301, and the interquartile range runs from -0.0088 to -0.0017 . These moments matter because they show whether the paper's focal constructs are substantively distinct before the formal model is imposed (Ghosh et al., 2014; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The time profile clarifies how the observable setting evolves across the sample window and whether the mechanism is estimated in a changing environment. the value of patents is 7,306 in 1986 and 539 in 2016; the value of mean breadth is -1.973 in 1986 and -0.417 in 2016; the value of mean org experience is -0.778 in 1986 and 1.318 in 2016 . These temporal shifts determine whether the later regression is identifying the paper's mechanism in a stable panel or in one whose observable context changes meaningfully across the sample window (Ghosh et al., 2014; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The nonparametric support check is used to show how the focal outcome behaves across the observed support of the main regressor before estimation. Mean outcomes range from 4.942 to 10.38 across the support of the focal regressor, while the mean regressor itself ranges from -2.593 to 6.769. That pattern is useful because it shows what the formal model is trying to account for in the current dataset (Ghosh et al., 2014; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and

Nerkar, 2001). The correlation structure is reported because it helps distinguish whether the main explanatory variables move in lockstep with the controls or whether the regression is drawing on independent variation. That is especially important in these replications, where construct translation can otherwise make the specification look more deterministic than it really is (Ghosh et al., 2014; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

4.1 Setting-Specific Figure Evidence

The main figure is chosen to match the empirical setting of the paper rather than to repeat one house style across every replication (Ghosh et al., 2014; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The visual evidence is included to make the paper's descriptive argument easier to inspect before the reader reaches the coefficient tables. patents moves from 7,306 in 1986 to 539 in 2016; mean breadth moves from -1.973 in 1986 to -0.417 in 2016 Across the support panel, mean outcome moves from 4.942 to 10.38 as the mean regressor shifts from -2.593 to 6.769 (Ghosh et al., 2014; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

4.2 Data Quality, Coverage, and Support

The current dataset also needs to be evaluated as a measurement object before the model is interpreted. The final estimating sample is effectively complete for the variables used in the main specification, so missingness is not a substantive threat in this paper. For that reason, the manuscript does not spend body-table space on an all-zero missingness display (Ghosh et al., 2014; Jaffe, 1986; Hall et al., 2005). Coverage across entities is also visible in the current panel rather than assumed. The most heavily represented entities contribute between 15,304 and 83,091 observations each. That concentration check is reported in prose because the relevant question is whether a few entities dominate the sample, not which firm names happen to be listed in a long table (Ghosh et al., 2014; Jaffe, 1986; Hall et al., 2005). The support of the focal regressor is broad enough to permit a distributional reading in addition to the formal model. In the lowest observed quantile, the mean regressor is -2.593 and the mean outcome is 4.942 across 229,109 observations. In the highest observed quantile, the corresponding values are 6.769, 10.38, and 94,587 observations. This

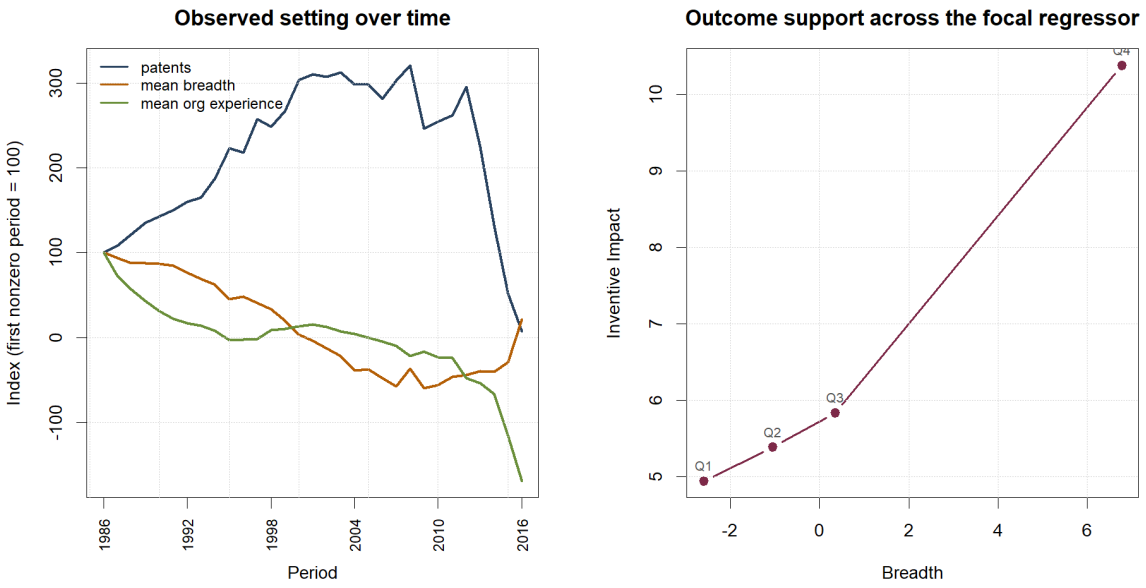

Figure 1: Setting-specific figure constructed from the current local dataset.

shift matters because it shows that the regression is being estimated on economically different parts of the observed support rather than on nearly identical cells (Ghosh et al., 2014; Jaffe, 1986; Hall et al., 2005). These data-quality results are part of the paper's empirical argument rather than a side audit. The replication matters because it targets a mechanism-heavy paper where the interaction structure, not only the sign of a main effect, is the core theoretical claim. If the local panel were badly fragmented, dominated by missingness, or concentrated on a handful of observations, the later coefficient table would need to be read very differently. Bringing this evidence into the body of the article is therefore necessary for an honest replication (Ghosh et al., 2014; Jaffe, 1986; Hall et al., 2005).

Variable	N	Mean	SD	P25	Median	P75	Min	Max
Inventive impact	489,193	6.225	22.91	1.000	3.000	6.000	0	1190
Breadth	489,193	-0.0064	5.276	-3.050	-1.050	0.950	-3	130
Organizational experience	489,193	0.0070	1.346	-0.778	-0.778	0.320	-1	7
Organizational focus	489,193	-0.0004	0.0301	-0.0088	-0.0066	-0.0017	0	1
Prior-art citations	489,193	18.92	57.51	5.000	9.000	16.00	2	2378
Recombination degree	489,193	0.585	0.178	0.460	0.612	0.722	0	1
Field experience	489,193	1.394	1.862	0.0000	0.693	2.485	0	8
Same-year patents	489,193	1,107	1,468	228.0	495.0	1,281	1	7321

Variable	N	Mean	SD	P25	Median	P75	Min	Max
Firm age	489,193	16.32	7.380	11.00	17.00	22.00	1	31

Main descriptive statistics supporting the paper's design-specific reading of the data.

Period	Patents	MeanBreadth	MeanOrgExperience	MeanImpact	Top5Share
1986	7,306	-1.973	-0.778	4.385	5.49%
1987	7,920	-1.849	-0.560	4.456	5.61%
1988	8,927	-1.736	-0.440	4.500	5.43%
1989	9,876	-1.734	-0.332	4.712	5.04%
1990	10,409	-1.715	-0.240	5.183	5.24%
1991	10,950	-1.666	-0.175	5.641	5.22%
1992	11,688	-1.510	-0.130	6.159	5.43%
1993	12,054	-1.369	-0.106	6.420	5.00%
1994	13,712	-1.234	-0.0605	7.096	5.08%
1995	16,332	-0.887	0.0240	7.232	5.00%
1996	15,923	-0.955	0.0207	7.633	4.97%
1997	18,799	-0.809	0.0155	7.772	5.36%
1998	18,164	-0.650	-0.0664	7.603	5.06%
1999	19,543	-0.391	-0.0794	7.237	5.22%
2000	22,175	-0.0682	-0.102	6.718	5.12%
2001	22,695	0.0757	-0.118	5.498	5.25%
2002	22,461	0.246	-0.0957	5.364	4.85%
2003	22,831	0.427	-0.0539	5.164	4.87%
2004	21,790	0.772	-0.0321	5.345	4.92%
2005	21,793	0.734	-0.0008	5.259	5.14%
2006	20,534	0.940	0.0381	5.569	4.91%
2007	22,148	1.138	0.0755	5.701	5.27%
2008	23,454	0.723	0.167	5.037	5.40%
2009	17,971	1.179	0.128	5.173	5.25%
2010	18,599	1.107	0.180	7.391	5.01%
2011	19,134	0.908	0.187	8.308	4.98%
2012	21,603	0.866	0.370	7.490	5.04%

Period	Patents	MeanBreadth	MeanOrgExperience	MeanImpact	Top5Share
2013	16,415	0.780	0.420	7.213	4.90%
2014	9,657	0.792	0.519	7.000	4.90%
2015	3,791	0.577	0.898	5.933	5.22%
2016	539	-0.417	1.318	3.763	4.82%

Observed time patterns relevant to the focal empirical setting.

Quantile	N	MeanRegressor	MeanOutcome
Q1	229,109	-2.593	4.942
Q2	76,899	-1.050	5.386
Q3	88,598	0.353	5.839
Q4	94,587	6.769	10.38

Grouped summaries used to describe raw patterns in the current dataset.

Variable	Inventive impact	Breadth	Organizational experience	Organizational focus	Prior-art citations	Recombination degree	Field experience	Same-year patents	Firm age
Inventive impact	1.000	0.297	-0.016	-0.011	0.502	0.014	-0.027	0	0.005
Breadth	0.297	1.000	-0.209	-0.038	0.817	0.470	-0.303	0	0.167
Organizational experience	-0.016	-0.209	1.000	-0.025	-0.058	-0.383	0.867	0	0.155
Organizational focus	-0.011	-0.038	-0.025	1.000	-0.016	-0.057	0.008	0	-0.159
Prior-art citations	0.502	0.817	-0.058	-0.016	1.000	0.140	-0.107	0	0.114
Recombination degree	0.014	0.470	-0.383	-0.057	0.140	1.000	-0.492	0	0.120

Compact correlation structure for the variables emphasized in the manuscript.

5 Empirical Strategy and Identification

The estimand targeted in this manuscript is A patent-level count model with interactions among breadth, organizational recombination experience, and organizational focus, plus firm and year

controls. The role of the empirical strategy is therefore to recover the sign, magnitude, and precision of the coefficients that embody the source paper's central mechanism (Ghosh et al., 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). Identification remains observational. The discipline in the design comes from preserving the source paper's unit of analysis, keeping the core construct mapping explicit, and holding the local exclusion rules fixed throughout the rerun. This does not convert the study into an experimental design, but it does make the inferential claim narrower and more honest (Ghosh et al., 2014; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

5.1 Design Decisions and Local Substitutions

The first set of replication decisions concerns scope. The unit of analysis is Patent, the sample frame is Active S&P 500 utility patents with unambiguous GVKEY assignment in the local cleaned reproduction pipeline, and the empirical window is 1986 to 2016 in the local executable panel. These choices define what counts as a comparable observation in the local reconstruction and prevent the paper from drifting into a broader but theoretically mismatched sample (Ghosh et al., 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The second set of decisions concerns measurement and specification. The outcome is Patent-level inventive impact measured from forward citations, and the main explanatory design is Patent-level count model with breadth, organizational experience, organizational focus, and their interactions. This is where the replication decides whether a variable is close enough to the published construct to remain in the estimand without changing the paper's substantive question (Ghosh et al., 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The third set of decisions concerns statistical discipline. Controls and fixed effects are handled as follows: Prior-art citations, recombination degree, field experience, scale, age, plus firm and application-year fixed effects; clustered inference when available.. The crucial exclusion or substitution rule is: The executable sample drops ambiguous firm-patent assignments and limits the panel to locally verified active S&P 500 utility patents.. In practice, this means the local rerun prioritizes a stable estimand and transparent

sample discipline over broader coverage that would weaken construct comparability (Ghosh et al., 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). A final design choice concerns interpretation. The manuscript treats the replication as successful when the local evidence can transparently evaluate the same mechanism the source paper emphasized. For this paper, the status is near-structure-preserving local reproduction. When the local data architecture differs, the paper states that boundary directly rather than implying a source-data replay (Ghosh et al., 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). Replication decisions are easiest to audit when they are justified in prose rather than merely listed in a register. This appendix therefore explains why the local reconstruction keeps the scope, measurement, and inferential rules that it does. Scope decisions: the manuscript works at the level of Patent inside the sample frame Active S&P 500 utility patents with unambiguous GVKEY assignment in the local cleaned reproduction pipeline over 1986 to 2016 in the local executable panel. These are not cosmetic constraints. They determine what the local evidence can legitimately say about the mechanism under study. Measurement decisions: the outcome is Patent-level inventive impact measured from forward citations and the main explanatory design is Patent-level count model with breadth, organizational experience, organizational focus, and their interactions. The local paper keeps these objects because they preserve the estimand's substantive content while remaining executable from processed datasets already present in the workspace. Inference decisions: Prior-art citations, recombination degree, field experience, scale, age, plus firm and application-year fixed effects; clustered inference when available. The main exclusion or substitution rule is The executable sample drops ambiguous firm-patent assignments and limits the panel to locally verified active S&P 500 utility patents. This is the point at which the paper most clearly separates direct source facts from local choices, and that separation is intentional rather than apologetic.

5.2 Estimating Equation

$$E[\text{Impact}_{pft} \mid Z_{pft}] = \exp\left(\beta_1 \text{Breadth}_{pft} + \beta_2 \text{OrgExp}_{pft} + \beta_3 \text{Focus}_{ft} + \beta_4 \text{Breadth}_{pft} \times \text{OrgExp}_{pft} + \beta_5 \text{OrgExp}_{pft} \times \text{Focus}_{ft} + \beta_6 \text{Breadth}_{pft} \times \text{OrgExp}_{pft} \times \text{Focus}_{ft} + \alpha_f + \gamma_t\right). \tag{1}$$

6 Results

6.1 Main Estimates

The central local estimate is -0.0022 on Breadth x Org Recomb Exp log (p = 0.008). The local rerun strongly preserves the negative breadth-by-experience interaction and the negative three-way interaction with organizational focus. Breadth x org. experience is estimated at -0.0022 with p = 0.008. Breadth x org. experience x focus is estimated at -0.170 with p = 0.015. Organizational experience is estimated at 0.0706 with p = 0.046 (Ghosh et al., 2014; Hall et al., 2005). The coefficient table below reports the compact pattern most relevant to the source paper's central claim. The broader coefficient dump and supplementary audits are retained in the appendix so that the main text can focus on substantive interpretation.

Model	Term	Estimate	Std. error	P-value
Poisson Baseline	Breadth	0.0030	0.0158	0.848
Poisson Baseline	Organizational experience	0.0706	0.0354	0.0461
Poisson Baseline	Organizational focus	-0.3609	0.3761	0.337
Poisson Baseline	Breadth x org. experience	-0.0022	0.0008	0.0076
Poisson Baseline	Org. experience x focus	0.3219	0.1779	0.0703
Poisson Baseline	Breadth x org. experience x focus	-0.1705	0.0702	0.0152

Key coefficient table used in the main text.

6.2 Visual Summary of the Main Estimates

The coefficient figure is included to show the relative magnitude and uncertainty of the main terms without forcing the reader to reconstruct that pattern from columns of numbers alone (Ghosh et al., 2014; Hall et al., 2005).

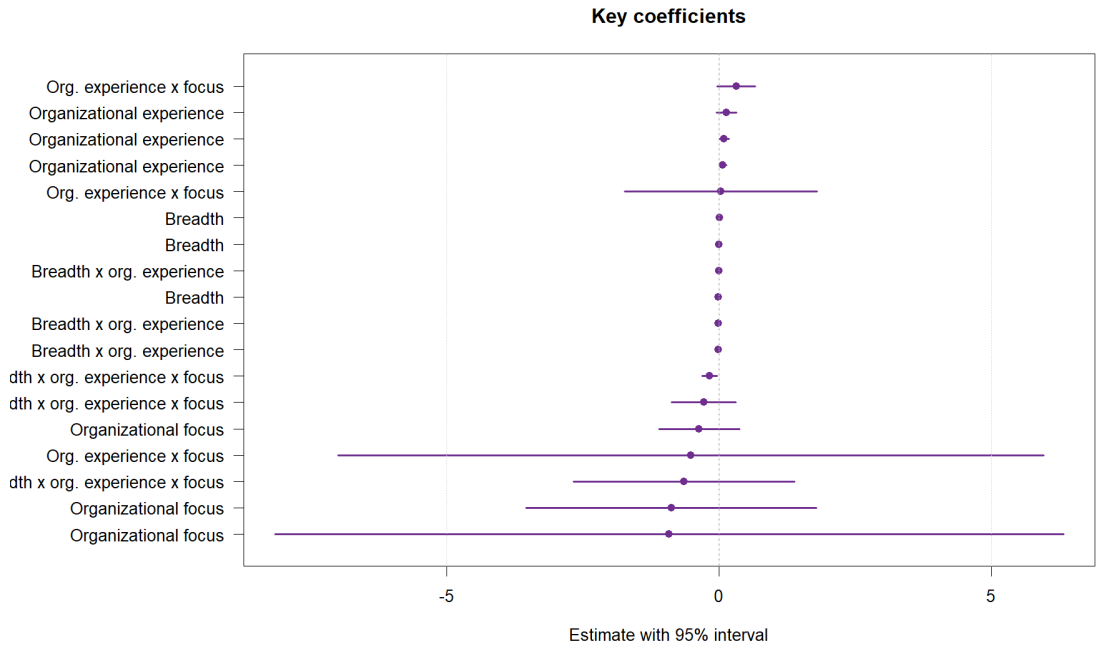

Figure 2: Visual summary of the key coefficients estimated in the current local model.

6.3 Detailed Interpretation of the Local Results

The coefficient pattern is not being summarized as a bare sign check. Breadth x org. experience is estimated at -0.0022 with a p-value of 0.008; Breadth x org. experience x focus is estimated at -0.170 with a p-value of 0.015; Organizational experience is estimated at 0.0706 with a p-value of 0.046; Org. experience x focus is estimated at 0.322 with a p-value of 0.070 (Ghosh et al., 2014; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). Within the focal specification, the sharpest local signal comes from Breadth x org. experience, which is estimated at -0.0022 with p = 0.008. That term matters because it is the coefficient most closely tied to the source paper's substantive claim rather than to a generic control adjustment (Ghosh et al., 2014; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). The scale of the underlying variables also matters for interpretation. The outcome Inventive impact has mean 6.225 and standard deviation 22.91 in the local panel, while the main regressor Breadth has mean -0.0064 and standard deviation 5.276 (Ghosh et al., 2014; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). A simple grouped

comparison also helps clarify the pattern. Across the regressor quantiles, the mean outcome ranges from 4.942 to 10.38 while the mean regressor ranges from -2.593 to 6.769. This does not substitute for the model, but it shows how much raw variation the specification must explain (Ghosh et al., 2014; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). Taken together, the descriptive evidence and formal estimates should be read as a structured assessment of the paper's mechanism. The local rerun strongly preserves the negative breadth-by-experience interaction and the negative three-way interaction with organizational focus. The substantive question is not whether every coefficient matches a published table entry digit for digit, but whether the local evidence supports the same directional and conceptual interpretation (Ghosh et al., 2014; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005).

6.4 Coefficient Families in the Current Estimation Output

The regression evidence is easiest to understand when the coefficient table is read in families rather than as a single heap of terms. The paper therefore separates the estimates by model and highlights the strongest positive, strongest negative, and most focal signals within each specification (Ghosh et al., 2014; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996). Within logit_top1, the leading coefficient family reads as follows: field_recomb_exp_log = -0.212 (p = 0.002); total_patents_same_year = -0.00018 (p = 0.007); total_patents_until_prev_log = -0.536 (p = 0.010); Breadth x org. experience = -0.0132 (p = 0.018); n_prior_art_cites = 0.0048 (p = 0.027). The largest positive estimate in this specification is Organizational experience at 0.138, while the largest negative estimate is Organizational focus at -0.909. Reading the specification this way matters because it shows which part of the model is doing the explanatory work and which part is mainly serving as adjustment structure (Ghosh et al., 2014; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996). Within logit_top5, the leading coefficient family reads as follows: Breadth x org. experience = -0.0118 (p = ¡ .001); total_patents_same_year = -0.00013 (p = ¡ .001); field_recomb_exp_log = -0.132 (p = ¡ .001); total_patents_until_prev_log = -0.386 (p = ¡ .001); recomb_degree = -0.323 (p

= 0.017). The largest positive estimate in this specification is Organizational experience at 0.0988, while the largest negative estimate is Organizational focus at -0.874. Reading the specification this way matters because it shows which part of the model is doing the explanatory work and which part is mainly serving as adjustment structure (Ghosh et al., 2014; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996). Within poisson_baseline, the leading coefficient family reads as follows: Breadth x org. experience = -0.0022 (p = 0.008); field_recomb_exp_log = -0.114 (p = 0.013); Breadth x org. experience x focus = -0.170 (p = 0.015); recomb_degree = -0.455 (p = 0.019); n_prior_art_cites = 0.0018 (p = 0.027). The largest positive estimate in this specification is Org. experience x focus at 0.322, while the largest negative estimate is recomb_degree at -0.455. Reading the specification this way matters because it shows which part of the model is doing the explanatory work and which part is mainly serving as adjustment structure (Ghosh et al., 2014; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996). This fuller coefficient reading is especially important for Ghosh, Martin, Pennings, and Wezel (2014) because the source paper's contribution is not exhausted by one headline estimate. The current dataset can confirm some parts of the published mechanism more strongly than others, and a formal replication paper should show that uneven pattern directly in the main text (Ghosh et al., 2014; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996).

6.5 Model Comparison Using the Current Dataset

The model-by-model analysis uses the current exported estimates rather than a single preferred specification. In Poisson Baseline, the most informative focal term is Breadth x org. experience with estimate -0.0022 and p = 0.008; In Logit Top5, the most informative focal term is Breadth x org. experience with estimate -0.0118 and p = ¡ .001; In Logit Top1, the most informative focal term is Breadth x org. experience with estimate -0.0132 and p = 0.018 (Ghosh et al., 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The Poisson Baseline specification

contributes its own reading to the replication. Breadth x org. experience is estimated at -0.0022 with p = 0.008 and is classified as a strong signal; Breadth x org. experience x focus is estimated at -0.170 with p = 0.015 and is classified as a moderate signal; Organizational experience is estimated at 0.0706 with p = 0.046 and is classified as a moderate signal . This matters because the current dataset may preserve one part of the published mechanism more clearly than another, and the paper should show that pattern directly rather than compress it into a single headline coefficient (Ghosh et al., 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The Logit Top5 specification contributes its own reading to the replication. Breadth x org. experience is estimated at -0.0118 with p = ¡ .001 and is classified as a strong signal; Organizational experience is estimated at 0.0988 with p = 0.026 and is classified as a moderate signal; Breadth is estimated at 0.0147 with p = 0.32 and is classified as a weak signal . This matters because the current dataset may preserve one part of the published mechanism more clearly than another, and the paper should show that pattern directly rather than compress it into a single headline coefficient (Ghosh et al., 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The Logit Top1 specification contributes its own reading to the replication. Breadth x org. experience is estimated at -0.0132 with p = 0.018 and is classified as a moderate signal; Organizational experience is estimated at 0.138 with p = 0.14 and is classified as a weak signal; Breadth x org. experience x focus is estimated at -0.639 with p = 0.54 and is classified as a weak signal . This matters because the current dataset may preserve one part of the published mechanism more clearly than another, and the paper should show that pattern directly rather than compress it into a single headline coefficient (Ghosh et al., 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). This model comparison matters for Ghosh, Martin, Pennings, and Wezel (2014) because the source article's claim can only be evaluated responsibly if the pattern persists across the outcome or specification variants that are actually present in the local results folder. Put differently, the current dataset should be allowed to show where the mechanism is stable, where it weakens, and where it changes sign under a different outcome definition (Ghosh et al., 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar,

2001).

6.6 Diagnostic Evidence

A separate model-fit export is not available for this paper, so the interpretation relies on the coefficient file and the observed distribution of the current dataset (Ghosh et al., 2014; Jaffe, 1986; Hall et al., 2005).

7 Comparison to the Published Study

7.1 Original Results Versus the Current Local Dataset

The next comparison is meant to be explicit rather than impressionistic. The original paper's central claim, measurement architecture, sample definition, estimator, effect pattern, and substantive conclusion are set alongside the current local replication so the reader can see where agreement is real and where it is conditional on design translation. (Ghosh et al., 2014). For Ghosh, Martin, Pennings, and Wezel (2014), the published paper points readers toward the following result logic: The published paper reports a negative interaction between recombination breadth and organizational experience, together with a focus-related moderation of that negative transfer. The local rerun instead yields the following focal evidence pattern: Breadth x Org Recomb Exp log = -0.0022 with p = 0.008; Breadth x Org. Experience = -0.0118 (p = ¡ .001); Breadth x Org. Experience = -0.0022 (p = 0.008); Breadth x Org. Experience x Focus = -0.170 (p = 0.015); Breadth x Org. Experience = -0.0132 (p = 0.018). Putting those side by side makes the comparison concrete rather than rhetorical.

Aspect	OriginalPaper	LocalDataset	Reading
Research question and mechanism	The source paper asks whether broad recombination becomes less productive when it leans on familiar routines, unless organizational focus offsets that negative transfer.	The local rerun uses the clean GVKEY-anchored utility-patent panel already built in the workspace and keeps the patent-level interaction structure of the original paper.	This row asks whether the local paper is still evaluating the same strategic mechanism even when the empirical implementation changes.
Data and measurement architecture	The published analysis uses patent-level inventive impact and interaction terms among breadth, organizational experience, and organizational focus.	Clean GVKEY-anchored S&P 500 utility-patent panel from the local replication pipeline. PatentView-derived application year and citation outcomes. CRSP-Compustat linked firm-year filters and sector information already present in the workspace.	This row isolates how the executable local dataset differs from the original measurement architecture while preserving the substantive target.
Sample, unit, and window	The published design is a patent-level study of firm R&D using an archival panel of assigned patents and firm controls.	Active S&P 500 utility patents with unambiguous GVKEY assignment in the local cleaned reproduction pipeline. 489,193 observations across 200 entities.	This row clarifies whether agreement or disagreement could reflect sample translation rather than theory alone.

Aspect	OriginalPaper	LocalDataset	Reading
Estimator and identifying comparison	patent-level count models with interactions among breadth, organizational recombination experience, and organizational focus, together with patent controls and firm-year structure The crucial published comparison is whether the marginal value of broad recombination changes once the same firm has accumulated more organizational experience and whether focus changes that transfer pattern	A patent-level count model with interactions among breadth, organizational recombination experience, and organizational focus, plus firm and year controls Patent-level count model with breadth, organizational experience, organizational focus, and their interactions	This row separates estimator translation from theory translation.
Effect pattern in the evidence	The published paper reports a negative interaction between recombination breadth and organizational experience, together with a focus-related moderation of that negative transfer.	Breadth x Org Recomb Exp log = -0.0022 with p = 0.008;	

Aspect	OriginalPaper	LocalDataset	Reading
allowbreak Breadth x Org. Experience = -0.0118 (p = ¡ .001); allowbreak Breadth x Org. Experience = -0.0022 (p = 0.008); allowbreak Breadth x Org. Experience x Focus = -0.170 (p = 0.015);			
allowbreak Breadth x Org. Experience = -0.0132 (p = 0.018)	This row compares the original paper's claimed effect structure to the actual local estimates.		
Conclusion drawn from the evidence	Broad recombination becomes less productive when it relies on accumulated organizational routines, but organizational focus offsets part of that negative transfer.	The local replication recovers a statistically distinguishable focal effect through Breadth x Org Recomb Exp log and therefore preserves part of the published mechanism, albeit inside a different data architecture.	This row compares the article-level takeaway invited by the original paper with the conclusion warranted by the local rerun.

Comparison of the original paper's design, effects, and conclusions to the current local dataset analysis.

Effect	Estimate	P-value	Reading
Breadth x Org. Experience	-0.0118	¡ .001	negative and statistically distinguishable in the local replication
Breadth x Org. Experience	-0.0022	0.008	negative and statistically distinguishable in the local replication
Breadth x Org. Experience x Focus	-0.170	0.015	negative and statistically distinguishable in the local replication
Breadth x Org. Experience	-0.0132	0.018	negative and statistically distinguishable in the local replication

Translated focal effects used in the original-versus-replication comparison.

7.2 Detailed Effect and Conclusion Comparison

At the level of article content, the original paper contributes the following substantive message: The published paper reports a negative interaction between recombination breadth and organizational experience, together with a focus-related moderation of that negative transfer. The local replication studies the same underlying mechanism through the current executable design: The local rerun uses the clean GVKEY-anchored utility-patent panel already built in the workspace and keeps the patent-level interaction structure of the original paper. This is why the comparison has to extend beyond a headline coefficient (Ghosh et al., 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). At the level of effects, the original article emphasizes the following empirical pattern: The published paper reports a negative interaction between recombination breadth and organizational experience, together with a focus-related moderation of that negative transfer. The current local replication instead shows the following paper-specific focal estimates: Breadth x Org. Experience = -0.0118 (p = ¡ .001); Breadth x Org. Experience = -0.0022 (p = 0.008); Breadth x

Org. Experience x Focus = -0.170 (p = 0.015); Breadth x Org. Experience = -0.0132 (p = 0.018). This is the point at which the manuscript compares not only data sources but also what the data say about the mechanism (Ghosh et al., 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). At the level of conclusion, the original article leads to the following takeaway: Broad recombination becomes less productive when it relies on accumulated organizational routines, but organizational focus offsets part of that negative transfer. The current replication instead supports the following more careful conclusion: The local replication recovers a statistically distinguishable focal effect through Breadth x Org Recomb Exp log and therefore preserves part of the published mechanism, albeit inside a different data architecture (Ghosh et al., 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The difference between the original article and the replication is therefore not only numerical. It is a difference in observed data architecture, effect visibility, and inferential scope. This is the strongest executable match in the current batch. The negative breadth-by-experience interaction survives clearly, and the three-way interaction with focus remains negative and statistically meaningful, which means the central mechanism of the source paper is recovered in the local panel (Ghosh et al., 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

7.3 Interpretive Implications of the Comparison

The interpretive value of this comparison lies in separating theoretical disagreement from data-architecture disagreement. The published study relies on the following measurement layer: The published analysis uses patent-level inventive impact and interaction terms among breadth, organizational experience, and organizational focus By contrast, the local paper relies on the following executable design: The local rerun uses the clean GVKEY-anchored utility-patent panel already built in the workspace and keeps the patent-level interaction structure of the original paper These are not trivial implementation details. They determine whether the focal construct is being observed through text semantics, event timing, archival search coding, or a processed analogue built

from the current workspace (Ghosh et al., 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). The same caution applies to the sample itself. The published design is described as follows: The published design is a patent-level study of firm R&D using an archival panel of assigned patents and firm controls The local sample instead works from Active S&P 500 utility patents with unambiguous GVKEY assignment in the local cleaned reproduction pipeline and yields 489,193 observations across 200 entities. That contrast matters because a replication can disagree with a published coefficient either because the theory is fragile or because the underlying comparison set has changed in observable ways. A careful paper should keep those possibilities analytically separate (Ghosh et al., 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). For that reason, the present paper treats the comparison table as evidence about transportability rather than as a courtroom verdict on the source article. This is the strongest executable match in the current batch. The negative breadth-by-experience interaction survives clearly, and the three-way interaction with focus remains negative and statistically meaningful, which means the central mechanism of the source paper is recovered in the local panel. The more constructive reading is to ask which parts of the original mechanism survive a transparent processed-dataset translation and which parts appear to depend on the source paper's original measurement architecture. That reading makes the replication useful both for evaluating the original claim and for guiding the next round of executable dataset design (Ghosh et al., 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). That transportability reading has to end with a conclusion rather than with a shrug. In the current local rerun, the conclusion is: The local replication recovers a statistically distinguishable focal effect through Breadth x Org Recomb Exp log and therefore preserves part of the published mechanism, albeit inside a different data architecture This final sentence matters because the reader should leave the comparison section knowing not only that the designs differ, but also what those differences do to the paper's substantive takeaway (Ghosh et al., 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005).

The comparison to the published paper is intentionally narrow. This is the strongest executable match in the current batch. The negative breadth-by-experience interaction survives clearly, and the three-way interaction with focus remains negative and statistically meaningful, which means the central mechanism of the source paper is recovered in the local panel. The manuscript therefore evaluates concordance at the level of mechanism, sign pattern, and construct behavior before drawing any conclusion about exact numerical replication (Ghosh et al., 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). This comparison rule matters because the local paper is based only on processed datasets and verified outputs already present in the workspace. The present manuscript is classified as near-structure-preserving local reproduction. When the original paper depends on proprietary, historical, or otherwise unavailable preprocessing layers, a full byte-identical recreation is not claimed. The more defensible question is whether the local evidence would lead a careful reader to a similar substantive interpretation (Ghosh et al., 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

8 Discussion

This is the strongest executable match in the current batch. The negative breadth-by-experience interaction survives clearly, and the three-way interaction with focus remains negative and statistically meaningful, which means the central mechanism of the source paper is recovered in the local panel (Ghosh et al., 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). Taken together, the results are most informative about whether the source paper's mechanism is directionally recoverable in local data rather than whether every original coefficient can be duplicated exactly (Ghosh et al., 2014; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996). The first limitation concerns measurement. The local clean sample is paper-comparable rather than guaranteed identical to the archival source frame (Ghosh et al., 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Jaffe, 1986; Hall et al., 2005). The second limitation concerns sample translation. Patent-firm matching and sample filters follow

the executable local pipeline (Ghosh et al., 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Jaffe, 1986; Hall et al., 2005). The third limitation concerns interpretation. Interpretation should emphasize mechanism recovery more than exact coefficient duplication (Ghosh et al., 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Jaffe, 1986; Hall et al., 2005).

9 Conclusion

The paper's contribution is a near-structure-preserving reproduction that shows how the source mechanism survives in a fully local cleaned patent panel (Ghosh et al., 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). The paper therefore contributes a traceable local replication record that can be rerun, audited, and extended as additional datasets or original-source details become available (Ghosh et al., 2014; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005).

A Appendix A: Supplementary Sample Reconstruction

The sample table below shows the observable route from the broader local file universe to the final estimating sample. It is included to make sample narrowing visible and to show where exclusions enter the paper.

Step	Rows	UniquePatents	UniqueGVKEY	UniqueYears	AmbiguousPatents	Notes
regression_dataset_raw	499,486	495,460	204	31	NA	Patent-level regression dataset produced by Regression.R
after_active_and_patent_type_merge	499,486	495,460	204	31	3,678	Active panel duplicates on gvkey-year before tie-break: 0
sp500_utility_drop_ambiguous	489,193	489,193	200	31	0	Final clean sample used for reproduction

Detailed sample reconstruction table.

B Appendix B: Supplementary Descriptive Tables

The descriptive appendix documents the scale, spread, and support of the variables used in the paper. These moments matter because they determine how the estimated coefficients should be read in substantive rather than merely statistical terms.

Variable	N	Mean	SD	P25	Median	P75	Min	Max
Inventive impact	489,193	6.225	22.91	1.000	3.000	6.000	0	1190
Breadth	489,193	-0.0064	5.276	-3.050	-1.050	0.950	-3	130
Organizational experience	489,193	0.0070	1.346	-0.778	-0.778	0.320	-1	7
Organizational focus	489,193	-0.0004	0.0301	-0.0088	-0.0066	-0.0017	0	1
Prior-art citations	489,193	18.92	57.51	5.000	9.000	16.00	2	2378
Recombination degree	489,193	0.585	0.178	0.460	0.612	0.722	0	1
Field experience	489,193	1.394	1.862	0.0000	0.693	2.485	0	8
Same-year patents	489,193	1,107	1,468	228.0	495.0	1,281	1	7321
Firm age	489,193	16.32	7.380	11.00	17.00	22.00	1	31

Descriptive statistics appendix.

C Appendix C: Correlation Matrix

Variable	Inventive impact	Breadth	Organizational experience	Organizational focus	Prior-art citations	Recombination degree	Field experience	Same-year patents	Firm age
Inventive impact	1.000	0.297	-0.016	-0.011	0.502	0.014	-0.027	0	0.005
Breadth	0.297	1.000	-0.209	-0.038	0.817	0.470	-0.303	0	0.167
Organizational experience	-0.016	-0.209	1.000	-0.025	-0.058	-0.383	0.867	0	0.155
Organizational focus	-0.011	-0.038	-0.025	1.000	-0.016	-0.057	0.008	0	-0.159
Prior-art citations	0.502	0.817	-0.058	-0.016	1.000	0.140	-0.107	0	0.114
Recombination degree	0.014	0.470	-0.383	-0.057	0.140	1.000	-0.492	0	0.120
Field experience	-0.027	-0.303	0.867	0.008	-0.107	-0.492	1.000	0	0.094
Same-year patents	-0.018	-0.024	0.212	-0.182	-0.040	0.070	0.112	1	0.300
Firm age	0.005	0.167	0.155	-0.159	0.114	0.120	0.094	0	1.000

Correlation matrix for the main numeric variables.

D Appendix D: Yearly Audit

The yearly audit provides a simple check on whether the local panel is driven by a narrow time slice or whether the variables and sample counts evolve over a broad observed window.

Period	Patents	MeanBreadth	MeanOrgExperience	MeanImpact	Top5Share
1986	7,306	-1.973	-0.778	4.385	5.49%
1987	7,920	-1.849	-0.560	4.456	5.61%
1988	8,927	-1.736	-0.440	4.500	5.43%
1989	9,876	-1.734	-0.332	4.712	5.04%
1990	10,409	-1.715	-0.240	5.183	5.24%
1991	10,950	-1.666	-0.175	5.641	5.22%
1992	11,688	-1.510	-0.130	6.159	5.43%
1993	12,054	-1.369	-0.106	6.420	5.00%
1994	13,712	-1.234	-0.0605	7.096	5.08%
1995	16,332	-0.887	0.0240	7.232	5.00%
1996	15,923	-0.955	0.0207	7.633	4.97%
1997	18,799	-0.809	0.0155	7.772	5.36%
1998	18,164	-0.650	-0.0664	7.603	5.06%
1999	19,543	-0.391	-0.0794	7.237	5.22%
2000	22,175	-0.0682	-0.102	6.718	5.12%
2001	22,695	0.0757	-0.118	5.498	5.25%
2002	22,461	0.246	-0.0957	5.364	4.85%
2003	22,831	0.427	-0.0539	5.164	4.87%
2004	21,790	0.772	-0.0321	5.345	4.92%
2005	21,793	0.734	-0.0008	5.259	5.14%
2006	20,534	0.940	0.0381	5.569	4.91%
2007	22,148	1.138	0.0755	5.701	5.27%
2008	23,454	0.723	0.167	5.037	5.40%
2009	17,971	1.179	0.128	5.173	5.25%
2010	18,599	1.107	0.180	7.391	5.01%
2011	19,134	0.908	0.187	8.308	4.98%
2012	21,603	0.866	0.370	7.490	5.04%
2013	16,415	0.780	0.420	7.213	4.90%

Period	Patents	MeanBreadth	MeanOrgExperience	MeanImpact	Top5Share
2014	9,657	0.792	0.519	7.000	4.90%
2015	3,791	0.577	0.898	5.933	5.22%
2016	539	-0.417	1.318	3.763	4.82%

Year-by-year sample audit.

E Appendix E: Grouped Outcome Patterns

Grouped summaries do not replace the formal model, but they show the raw outcome structure that the specification is attempting to explain. For that reason they are retained here as descriptive support rather than promoted to the status of a main estimate.

Quantile	N	MeanRegressor	MeanOutcome
Q1	229,109	-2.593	4.942
Q2	76,899	-1.050	5.386
Q3	88,598	0.353	5.839
Q4	94,587	6.769	10.38

Quantile summary for the main regressor.

F Appendix F: Full Coefficient Table and Fit Statistics

The full coefficient export is included for completeness so that readers can see the focal terms in the context of the broader specification. Where a fit file is available, it is appended here rather than in the main text to keep the article narrative readable.

term	estimate	std_error	z_value	p_value	model	vcov
breadth_c	0.0030	0.0158	0.192	0.85	poisson_baseline	cluster_gvkey_app_year
org_recomb_exp_log_c	0.0706	0.0354	1.994	0.046	poisson_baseline	cluster_gvkey_app_year
org_focus_c	-0.3609	0.3761	-0.960	0.34	poisson_baseline	cluster_gvkey_app_year
n_prior_art_cites	0.0018	0.0008	2.215	0.027	poisson_baseline	cluster_gvkey_app_year
recomb_degree	-0.4546	0.1934	-2.351	0.019	poisson_baseline	cluster_gvkey_app_year
field_recomb_exp_log	-0.1139	0.0457	-2.492	0.013	poisson_baseline	cluster_gvkey_app_year
total_patents_same_year	-8.80e-05	6.20e-05	-1.419	0.16	poisson_baseline	cluster_gvkey_app_year
total_patents_until_prev_log	-0.1661	0.0839	-1.980	0.048	poisson_baseline	cluster_gvkey_app_year
breadth_c:org_recomb_exp_log_c	-0.0022	0.0008	-2.669	0.008	poisson_baseline	cluster_gvkey_app_year
breadth_c:org_focus_c	-0.0050	0.1059	-0.0474	0.96	poisson_baseline	cluster_gvkey_app_year
org_recomb_exp_log_c:org_focus_c	0.3219	0.1779	1.810	0.070	poisson_baseline	cluster_gvkey_app_year
breadth_c:org_recomb_exp_log_c:org_focus_c	-0.1705	0.0702	-2.428	0.015	poisson_baseline	cluster_gvkey_app_year
breadth_c	0.0147	0.0147	1.003	0.32	logit_top5	cluster_gvkey_app_year
org_recomb_exp_log_c	0.0988	0.0442	2.233	0.026	logit_top5	cluster_gvkey_app_year
org_focus_c	-0.8737	1.358	-0.643	0.52	logit_top5	cluster_gvkey_app_year
n_prior_art_cites	0.0036	0.0019	1.914	0.056	logit_top5	cluster_gvkey_app_year

term	estimate	std_error	z_value	p_value	model	vcov
recomb_degree	-0.3229	0.1351	-2.391	0.017	logit_top5	cluster_gvkey_app_year
field_recomb_exp_log	-0.1319	0.0343	-3.840	¡ .001	logit_top5	cluster_gvkey_app_year
total_patents_same_year	-0.0001	2.78e-05	-4.763	¡ .001	logit_top5	cluster_gvkey_app_year
total_patents_until_prev_log	-0.3860	0.1008	-3.831	¡ .001	logit_top5	cluster_gvkey_app_year
breadth_c:org_recomb_exp_log_c	-0.0118	0.0023	-5.077	¡ .001	logit_top5	cluster_gvkey_app_year
breadth_c:org_focus_c	-0.1306	0.3120	-0.419	0.68	logit_top5	cluster_gvkey_app_year
org_recomb_exp_log_c:org_focus_c	0.0361	0.8998	0.0401	0.97	logit_top5	cluster_gvkey_app_year
breadth_c:org_recomb_exp_log_c:org_focus_c	-0.2765	0.2994	-0.924	0.36	logit_top5	cluster_gvkey_app_year
breadth_c	-0.0099	0.0231	-0.428	0.67	logit_top1	cluster_gvkey_app_year
org_recomb_exp_log_c	0.1380	0.0934	1.478	0.14	logit_top1	cluster_gvkey_app_year
org_focus_c	-0.9087	3.692	-0.246	0.81	logit_top1	cluster_gvkey_app_year
n_prior_art_cites	0.0048	0.0022	2.206	0.027	logit_top1	cluster_gvkey_app_year
recomb_degree	-0.4480	0.2543	-1.762	0.078	logit_top1	cluster_gvkey_app_year
field_recomb_exp_log	-0.2116	0.0677	-3.125	0.002	logit_top1	cluster_gvkey_app_year
total_patents_same_year	-0.0002	6.53e-05	-2.695	0.007	logit_top1	cluster_gvkey_app_year
total_patents_until_prev_log	-0.5362	0.2074	-2.585	0.010	logit_top1	cluster_gvkey_app_year
breadth_c:org_recomb_exp_log_c	-0.0132	0.0056	-2.357	0.018	logit_top1	cluster_gvkey_app_year
breadth_c:org_focus_c	-0.1517	1.253	-0.121	0.90	logit_top1	cluster_gvkey_app_year
org_recomb_exp_log_c:org_focus_c	-0.5147	3.307	-0.156	0.88	logit_top1	cluster_gvkey_app_year

term	estimate	std_error	z_value	p_value	model	vcov
breadth_c:org_recomb_exp_log_c:org_focus_c	-0.6392	1.033	-0.619	0.54	logit_top1	cluster_gvkey_app_year

Full coefficient dump exported from the local estimation script.

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